Healthcare Analysis

# Hospital Operations, Cost & Patient Flow Intelligence

Cost & encounter analysis: claim-cost variation and encounter duration


Presented by:

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Programme:

**Week 3 · Analystlab Africa Data science internship**

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# Research Question

## How do total claim costs vary across encounter classes, and is encounter duration associated with total claim cost?

**Decision supported:** Where hospital operations and finance teams should focus further investigation of encounter costs.

**Why it matters:** Cost patterns can reveal encounter classes that warrant closer financial review.

**Evidence boundary:** Historical, observational data can show patterns and associations. It cannot establish causation.

# Objectives

## 01 · Profile the Records

Inspect source tables, fields, relationships and data quality.

## 02 · Prepare Fields

Parse dates; calculate encounter duration; derive year, cost and coverage indicators.

## 03 · Explore Cost Patterns

Compare encounter volume, claim-cost distributions, duration and recorded payer coverage.

## 04 · Test Two Hypotheses

Assess cost differences across encounter classes and the association between duration and cost.

# Six Data Tables

**27,891** encounter records retained. Encounters are the primary analysis table; the other tables give context and checks.

- Encounters: primary analysis
- Procedures: cost reconciliation
- Payers: payer names
- Organizations: organisation check
- Patients: ID-only key check
- Data dictionary: field definitions

The notebook identifies a single organisation. It does not establish that findings generalise to other hospitals.

# Data Preparation Workflow

| | | |
|---|---|---|
| 1 | **Inspect** | Check fields, keys, duplicates, missingness and data types. |
| 2 | **Transform** | Parse START and STOP as UTC datetimes; calculate duration in hours and encounter year. |
| 3 | **Enrich** | Add zero-cost, extreme-duration, log-cost, payer-name and zero-coverage indicators. |
| 4 | **Validate** | Retain all 27,891 encounters; check unique IDs and required analysis fields. |

**PREPARATION DECISION**

**No rows were dropped and no missing values were imputed. Unusual values were flagged and retained for interpretation.**

# Data Quality Checks

## Keys & Duplicates

Foreign-key matches were reported at 100%. No duplicate encounter IDs; repeated patient/time/class combinations were retained as potentially distinct services.

## Missing Fields

Gaps occur in reason-code and reason-description fields. They were treated as “no reason recorded”, not automatically as errors.

## Dates & Duration

Dates parsed successfully. Some durations exceed one year; some encounter classes have fixed durations.

## Cost & Coverage

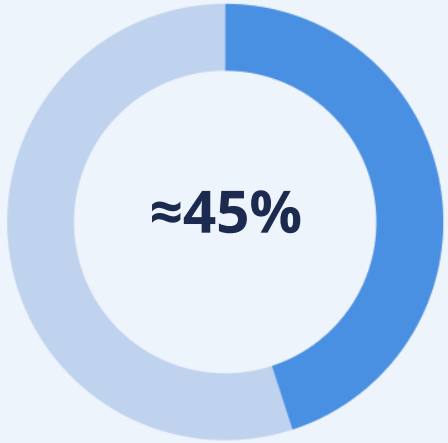
Some zero claim costs coexist with positive base costs. Zero payer coverage is ambiguous, including records with named insurers.

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# Encounter Volume

- Ambulatory: largest class, about 45% of records
- Inpatient: the smallest class
- Annual volume peaks in 2014 and 2021
- 2011 and 2022 are partial years; do not compare them with full years

## VOLUME PATTERN IN THE NOTEBOOK




Illustrative split based on the notebook's reported ~45% ambulatory share; not a full class distribution.

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# Skewed Claim Costs

- Long tail: a few expensive encounters pull the mean above the median
- Original scale compresses most records near zero; log scale reveals multiple peaks
- Report medians and quartiles alongside means
- Use log scaling to inspect the distribution visually
- Prefer rank-based tests over methods that assume normal costs

**Median ≠ full cost exposure**

# Class, Duration & Payer

### Cost by class

Classes occupy different cost levels. Some have wide within-class spreads; class alone does not explain why costs differ.

### Duration vs cost

Where duration varies, longer encounters tend to coincide with higher costs. Several classes have fixed durations, limiting interpretation.

### Payer coverage

Zero recorded coverage varies by class and payer. A named insurer can still appear with zero coverage; this is not equivalent to uninsured status.

Interpret carefully: procedures, payer and patient characteristics may relate to both encounter class and cost.

# Hypotheses & Rank-Based Tests

## H1 · Group differences

### Do claim-cost distributions differ across encounter classes?

$H_0$: Same distribution across classes.

$H_1$: At least one class differs.

Method: Kruskal–Wallis test; independent groups, non-normal cost distributions.

## H2 · Association

### Is encounter duration associated with claim cost?

$H_0$: Spearman $\rho = 0$.

$H_1$: A monotonic association exists.

Method: Spearman rank correlation; assesses monotonic association without requiring a straight-line relationship.

The notebook notes that repeated encounters may weaken independence for H1, and fixed-duration classes constrain H2.

# Results & Interpretation

| | | |
|---|---|---|
| 1 | **H1 · Cost differs by class** | The notebook's interpretation reports a statistically significant, moderate effect ($\varepsilon^2$) and pairwise comparisons with multiplicity correction. The conclusion survives a one-encounter-per-patient check. |
| 2 | **H2 · Duration and cost** | Spearman correlation and its p-value are calculated in the notebook. The written interpretation explains how to read the sign and magnitude, but the executed statistic is not embedded in the supplied file. |
| 3 | **Reporting limitation** | The notebook file contains analysis code and narrative but no saved cell outputs. Exact test statistics, p-values, confidence intervals and numeric effect size cannot be verified from this file alone. |

# Limitations & Recommendations

## Limitations

- Repeated patient encounters may violate independence
- Extreme and fixed durations constrain comparisons
- Skewed costs and ambiguous zero coverage
- Potential confounding by procedures, payer and patient mix
- One organisation and a historical period limit generalisability

## Recommendations

- Review high-median-cost classes and underlying services
- Report medians alongside averages
- Investigate high-cost encounters with procedure and patient context
- Clarify zero-coverage definitions with the data owner

Do not infer causality or change operations based on these patterns alone. Validate with richer clinical and financial data.

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# Thank You

This project moved from analysing descriptive patterns in data to asking better questions for operational review.

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